

Why These Rules of Immunity Matter

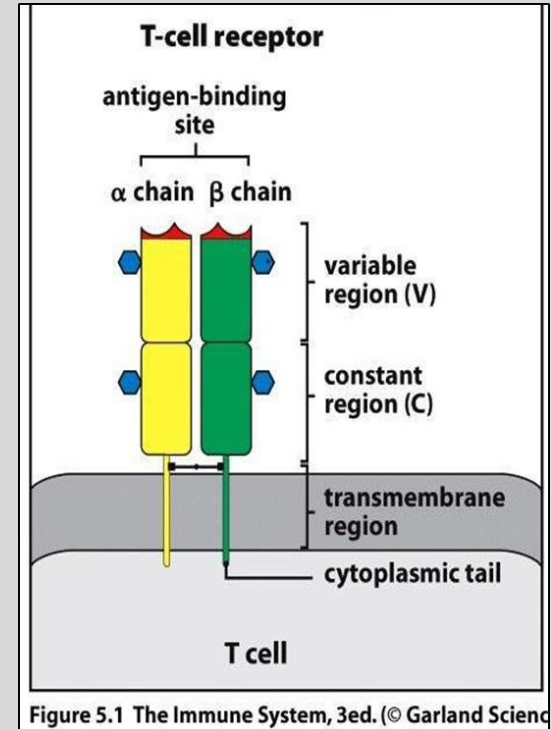
- These rules evolved to prevent inappropriate or harmful immune activation
- Adaptive immunity follows non-negotiable rules
- These rules determine what the immune system can and cannot respond to
- Understanding the problem explains the mechanism

Learning Objectives / Conceptual Focus

- Explain why T-Cells require antigen presentation
- Describe the role of MHC molecules in immune safety
- Connect antigen presentation to tolerance and disease
 - Emphasis on mechanisms rather than memorization

The Unique Role of T Cells

- Specialized for detecting intracellular threats
- Focus on viral infection and malignant transformation
- Function as inspectors rather than direct antigen binders
 - Requires intermediary systems to safely interpret antigen information



T-Cells Do Not Bind Free Antigen

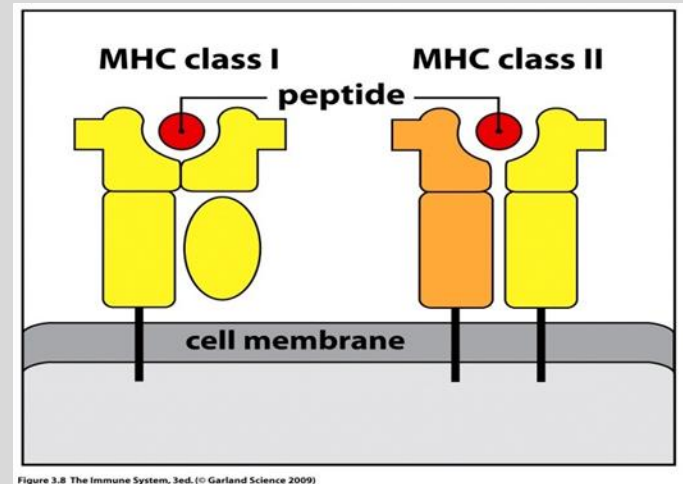
- Free antigen recognition would be unsafe
- Processing inserts contextual control
- Prevents responses to harmless exposure
 - Prevents activation based solely on molecular presence

The Major Histocompatibility Complex (MHC)

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- Forces T-Cell recognition to occur in a controlled cellular context
- Encodes the immune system's peptide display platform
- Known as HLA molecules in humans
- Enables standardized antigen reporting



Clinical Consequences of MHC Diversity

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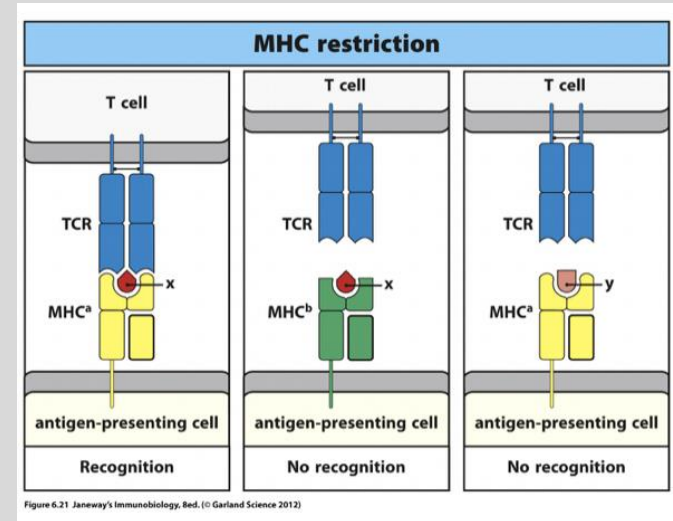
- Differences in disease susceptibility
 - Reflects differences in peptide presentation efficiency
- Variable responses to infection and vaccines
- Central importance in transplantation

MHC Restriction Defines T-Cell Recognition

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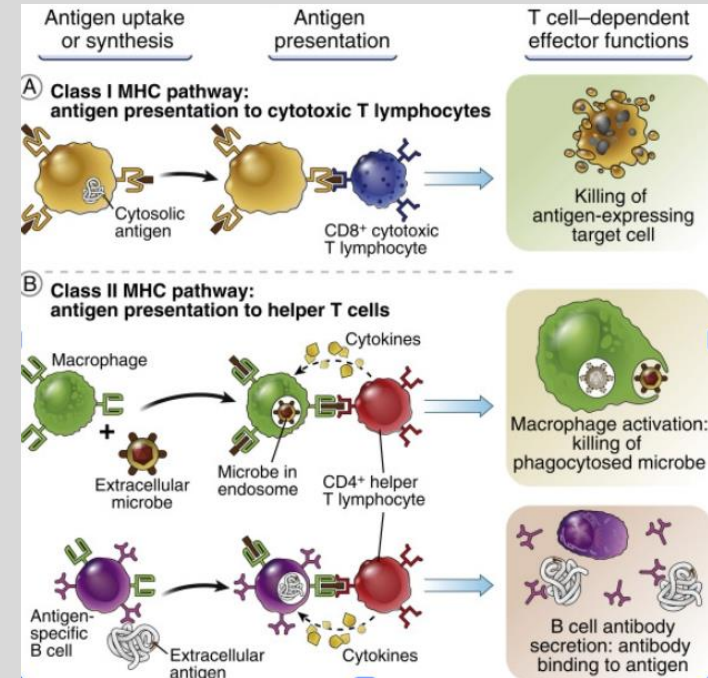
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- T cells recognize peptide plus MHC
- Specificity depends on MHC context
- Explains personalized T-cell responses
 - Same antigen may elicit different responses across individuals



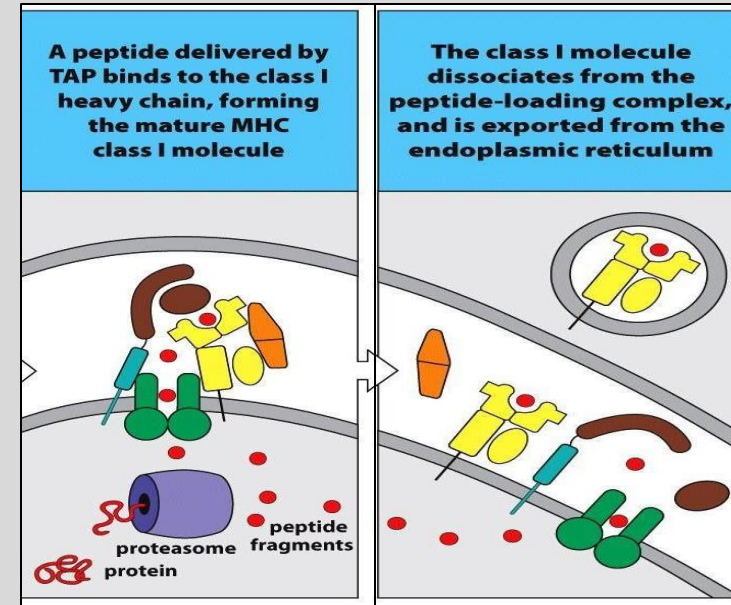
Why Two MHC Classes Exist

- Antigens arise from different cellular locations
- Intracellular vs extracellular threats
- Distinct immune responses required
 - Different threats demand different decision pathways



MHC Class I: Continuous Intracellular Surveillance

- Presents peptides from intracellular proteins
- Expressed on all nucleated cells
- Enables early detection of abnormal cells



Continuous Nature of Class I Presentation

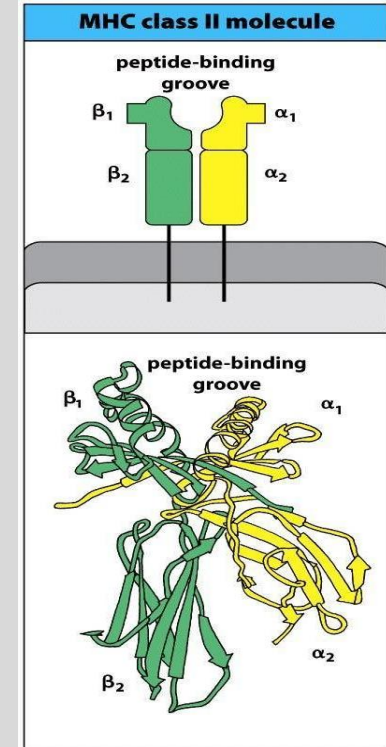
- Antigen reporting is non-optional
 - Absence of presentation itself can signal abnormality
- Cells cannot hide from surveillance
- Supports early, localized immune responses

Why a Second Presentation Pathway Is Needed

- Some immune decisions must be made at the systems level
 - Individual cell killing is insufficient for coordinated immune responses
- Extracellular threats require coordination
 - Centralizes immune activation authority
 - Antibody production and immune orchestration require helper T-Cell input
- Cytotoxicity alone is insufficient
- Leads to MHC class II pathway

Restriction of MHC Class II Expression

- Limited to professional APCs
- Ensures controlled immune activation
- Prevents widespread inappropriate responses



Targeting Class II Molecules to Endosomes

- Invariant chain directs trafficking
- Ensures exposure to extracellular peptides only
 - Spatial routing enforces correct antigen source
- Critical for specificity

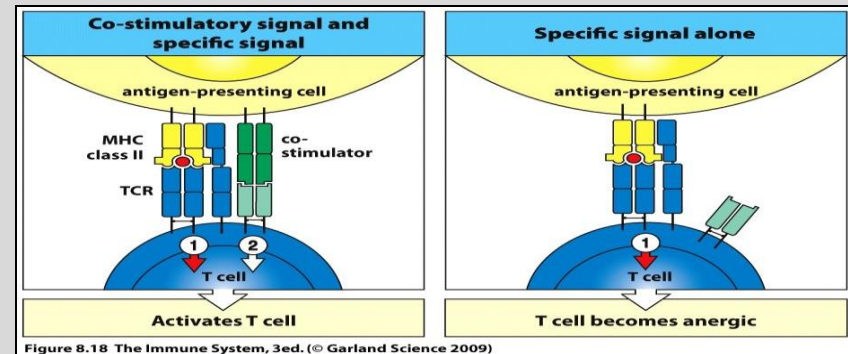
Antigen Presentation Is Not Enough

- Recognition alone does not activate T cells
 - Antigen recognition is separated from immune action by design
- Prevents responses to harmless antigens
- Separates recognition from action
 - Prevents immune responses to harmless or self antigens

Costimulation: The Second Signal

- Provides contextual information
 - Confirms that antigen recognition occurs in infection or tissue damage
- Indicates danger or inflammation
- Required for full T-cell activation

- Absence of costimulation leads to tolerance, not immunity



Why Only Professional APCs Activate Naïve T Cells

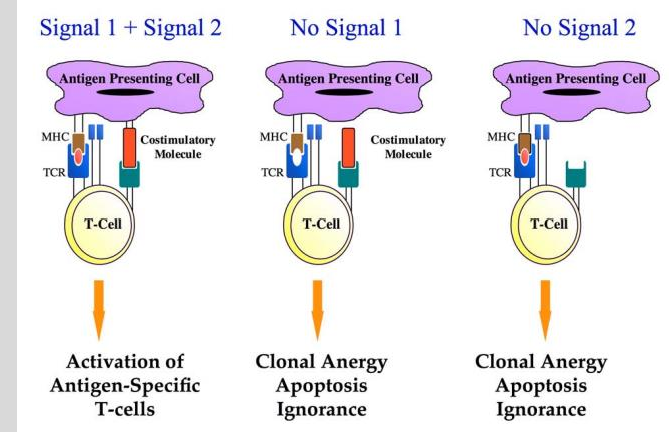
- Deliver both antigen and costimulation
- Centralizes immune decision-making
 - Prevents widespread, unregulated T-Cell activation
- Supports peripheral tolerance

Antigen Recognition Without Costimulation

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- Leads to functional unresponsiveness
- Known as anergy
- Protects against autoimmunity
 - Functionally silences potentially self-reactive T-Cells



Costimulation and Tolerance Are Linked

- Central tolerance is incomplete
- Peripheral mechanisms provide backup
- Costimulation requirements enforce restraint
 - T-Cells encountering antigen without signal 2 become functionally inactive
 - Failure to deliver signal 2 actively promotes tolerance
 - Peripheral tolerance is an intentional outcome, not an accident

Clinical Relevance

- Costimulatory pathways are important targets for clinical intervention
- Therapeutic targets exploit normal immune regulatory checkpoints
 - Checkpoint inhibitors modify costimulatory balance to enhance or suppress immunity
- Therapeutic manipulation can enhance or suppress immune responses

Integrating Antigen Presentation, Costimulation, and Tolerance

- Antigen presentation provides specificity
- Costimulation determines whether a response should occur
- Tolerance mechanisms prevent harmful self-reactivity

Superantigens: A Major Exception

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- Bypass normal antigen specificity
 - Violates normal rules of antigen specificity and safety
- Bind MHC and TCR externally
- Trigger widespread activation

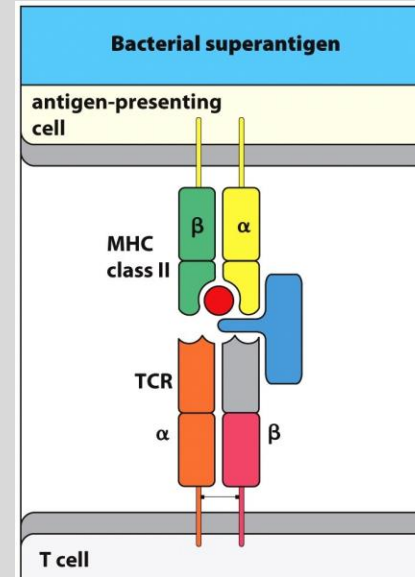
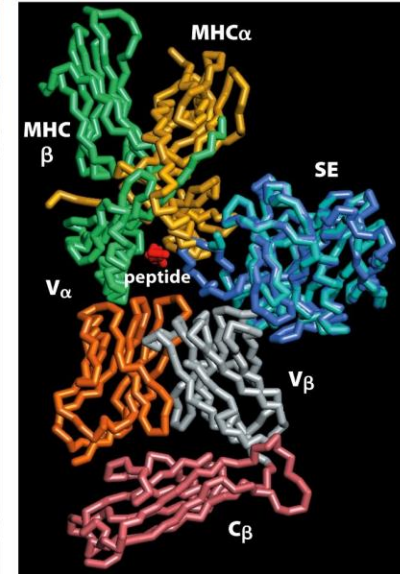


Figure 11.8 The Immune System, 3ed. (© Garland Science 2009)



Consequences of Superantigen Activation

- Massive cytokine release
- Systemic inflammation
- Severe clinical syndromes

Typical Response	Superantigen Response
Antigen must be processed by APC	APC Processing not required
Antigen recognition and T-Cell activation is MHC-II restricted	MHC-II cells required for activation by not MHC-II restricted
Small amount of T-Cells activated	20-30% T-Cell activated

What Superantigens Teach Us About Design

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- Normal rules exist for safety
 - Restrictions in immune recognition are intentional, not inefficient
- Shortcuts reveal system vulnerabilities
- Precision is essential

Genetic Variation and Disease Susceptibility

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- MHC alleles influence risk
- Affect peptide presentation efficiency
- Link genetics to immunopathology

Antigen Presentation in Transplantation

- Antigen Presentation is central to transplant rejection
- Recipient T Cells may recognize donor MHC molecules as foreign
- MHC matching is critical

Integrating Antigen Presentation and Regulation

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- Presentation provides specificity
- Costimulation provides context
- Tolerance provides restraint

Key Takeaways

- Antigen presentation determines immune visibility
- MHC constrains T-cell recognition
- Regulation ensures immune precision

Summary Slide

- **Summary Point 1**

- Antigen presentation determines **what the immune system can see**
- MHC molecules constrain T-cell recognition to specific, regulated contexts
- This design prevents indiscriminate or dangerous immune activation

- **Summary Point 2**

- Costimulation and tolerance mechanisms determine **whether a response occurs**
- Separation of recognition from action protects the host
- Failures in these controls explain autoimmunity, transplant rejection, and immune evasion

- Core References:

- Abbas AK, Lichtman AH, Pillai S. *Cellular and Molecular Immunology*. Elsevier.
- Murphy K, Weaver C. *Janeway's Immunobiology*. Garland Science.
- Parham P. *The Immune System*. Garland Science.
- Klein J, Sato A. "The HLA system." *New England Journal of Medicine*.